STABILITY EXERCISES:

1. A ship has a displacement of 3200 tonnes (KG 3 m and KM 5.5 m). She then loads 5200 tonnes of cargo (KG 5.2 m). Find how much deck cargo having a KG 10 m may now be loaded if the ship is to complete loading with a positive GM of 0.3 m
2. A ship of 3500 tonnes light displacement and light KG 6.4 m has to load 9600 tonnes of cargo. The KG of the lower hold is 4.5 m, and that of the tween deck is 9 m. The load KM is 6.2 m and, when loading is completed, the righting moment at 6 degrees of heel is required to be 425 tonnes.m. Calculate the amount of cargo to be loaded into the lower hold and tween deck, respectively.
3. An oil tanker 150 m long, displacement 12 500 tonnes, MCT 1 cm 200 tonnes.m, leaves port with drafts 7.2 m F and 7.4 m A. There is 550 tonnes of fuel oil in the foward deep tank (centre of gravity 70 m forward of the centre of flotation) and 600 tonnes in the after deep tank (centre of gravity 60 m aft of centre of flotation). The centre of flotation is 1 m aft of amidships. During the sea passage 450 tonnes of oil is consumed from aft. Find how much oil must be transferred from the forward tank to the after tank if the ship is to arrive on an even keel
4. A ship 100 m long, and with a displacement of 2200 tonnes, has longitudinal metacentric height 150 m. The present drafts are 5.2 m F and 5.3m A. Centre of flotation is 3 m aft of amidships. Find the new drafts if a weight of 5 tonnes already on board is shifted aft through a distance of 60 metres.
5. A ship of 5500 tonnes displacement has KG 5 m, and she proceeds to load the following cargo:
   1000 tonnes KG 6 m
   700 tonnes KG 4 m
   300 tonnes KG 5 m
   She then discharges 200 tonnes of ballast KG 0.5 m. Find how much deck cargo (KG 10 m) can be loaded so that the ship may sail with a positive GM of 0.3 metres. The load KM is 6.3 m.